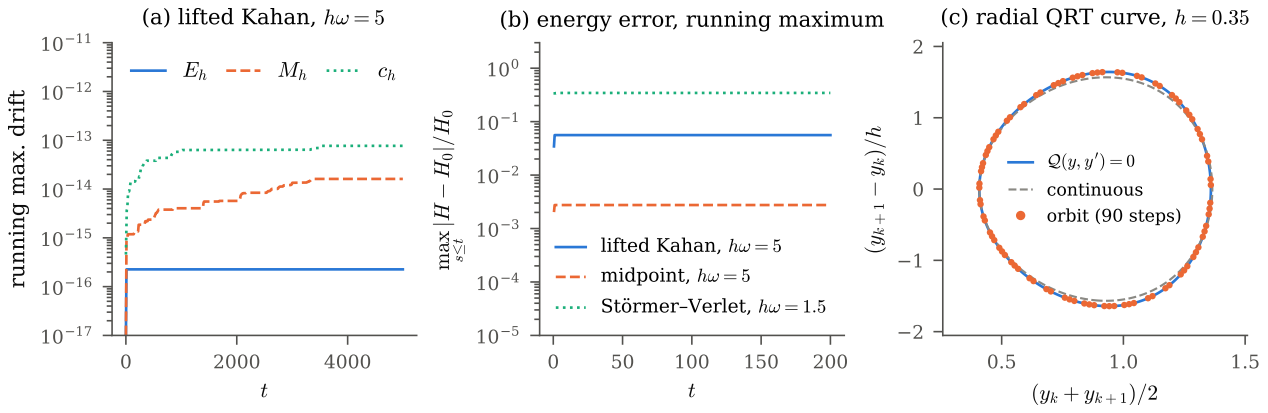

(a) lifted Kahan, $h\omega = 5$
running max. drift
E_h
M_h
c_h
t
(b) energy error, running maximum
$\max_{s \le t} |H - H_0|/H_0$
lifted Kahan, $h\omega = 5$
midpoint, $h\omega = 5$
Störmer–Verlet, $h\omega = 1.5$
t
(c) radial QRT curve, $h = 0.35$
$(y_{k+1} - y_k)/h$
$\mathcal{Q}(y, y') = 0$
continuous
orbit (90 steps)
$(y_k + y_{k+1})/2$